

Andi Zhang

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Education

University of Cambridge

PhD in Computer Science

Oct. 2019 – Jul. 2024

- Supervised by Dr. Damon Wischik, focused on Bayesian neural networks and probabilistic generative models, delved into applications related to Out-Of-Distribution (OOD) detection.

MASt in Mathematics (Part III)

Oct. 2018 – Jun. 2019

- Relevant modules: Statistical Learning in Practice, Convex Optimization, Astrostatistics, Bayesian Modelling and Computation, Modern Statistical Methods

MPhil in Advanced Computer Science

Oct. 2017 – Jun. 2018

- Relevant modules: Machine Learning for Natural Language Processing, Algebraic Path, Deep Learning for Natural Language Processing, Advanced Functional Programming, Probabilistic Machine Learning

University of Manchester

BSc (Hons) in Computer Science and Mathematics

Sep. 2014 – Jun. 2017

- Relevant modules: Machine Learning, Quantum Computing, Compilers, Convex Optimization, Probability, Statistical Methods, Functional Analysis, Topology, Number Theory

Experience

University College London

London, United Kingdom

Visitor | Mentor: Dr. David Barber

July. 2023 – Oct. 2023

- Working on a new image generative model that based on Diffusion models.
- Working on Amortized LoRA, a more principle and probabilistic way to generate LoRA parameters than Hyperdreambooth.

University of Cambridge

Cambridge, United Kingdom

Course Supervisor (Teaching Assistant)

Oct. 2019 – Present

- Supervised IA Algorithms (2020 Lent, 2023 Lent), IA Machine Learning in Real World (2020 Lent), IA Introduction to Probability (2020 Easter, 2021 Easter), IB Further Java (2020 Michaelmas), IB Data Science (2022 Michaelmas, 2023 Michaelmas), IB Further Graphics (2022 Michaelmas) and II Machine Learning and Bayesian Inference (2023 Lent)

MRC Biostatistics Unit, University of Cambridge

Cambridge, United Kingdom

Research Assistant | Supervisor: Dr. Sofia Villar and Dr. David Robertson

Jul. 2019 – Sep. 2019

- Developed a simulation system to tackle the multi-armed bandit problem (MABP) in clinical trial settings and pioneered a unique test statistic for binary response MABP situations. Although the time constraints prevented this work from being formalized into a publication, it provided substantial practical and theoretical insight.

Tencent (WeChat AI)

Beijing, China

R & D Intern | Mentor: Dr. Jie Zhou

Jul. 2018 – Sep. 2018

- Engaged in comprehensive exploration and research in deep reinforcement learning and evolution strategies. While this research did not culminate in a tangible output, the experience gained in these cutting-edge fields has proven invaluable.

Amazon (Alexa)

Cambridge, United Kingdom

Applied Scientist Intern | Mentor: Dr. Emilio Monti

Jun. 2017 – Sep. 2017

- Employed the Seq2Seq model, the predecessor of the Transformer, with the Attention mechanism for semantic parsing tasks. Semantic parsing involves translating natural language into a structured internal language used for database or knowledge base querying.

Robotics Institute, Carnegie Mellon University

Pittsburgh, United States

Student Intern | Supervisor: Dr. Stelian Coros

Jun. 2016 – Sep. 2016

- Contributed to the implementation of a general stance controller for 3D-printable robotic creatures. The controller's adaptability enables it to be used with any human-designed multi-foot robots, hence its designation as 'general'.

Publications

- Falsehoods that ML researchers believe about OOD detection** NeurIPS 2022 Workshop
Andi Zhang, Damon Wischik
In ML Safety Workshop, 36th Conference on Neural Information Processing Systems
- Out-of-Distribution Detection with Class Ratio Estimation** NeurIPS 2022 Workshop
Mingtian Zhang*, **Andi Zhang***, Tim Z. Xiao, Yitong Sun, Steven McDonagh
In ML Safety Workshop, 36th Conference on Neural Information Processing Systems
- On the Out-of-distribution Generalization of Probabilistic Image Modelling** NeurIPS 2021
Mingtian Zhang*, **Andi Zhang***, Steven McDonagh
In Neural Information Processing Systems annual meeting, 2021
- Measuring Uncertainty in Signal Fingerprinting with Gaussian Processes Going Deep** IPIN 2021
Ran Guan, **Andi Zhang**, Mengchao Li, Yongliang Wang
In International Conference on Indoor Positioning and Indoor Navigation, 2021
- Towards dynamic and scalable active learning with neural architecture adaption for object detection** BMVC 2021
Fuhui Tang, Dafeng Wei, Chenhan Jiang, Hang Xu, **Andi Zhang**, Wei Zhang, Hongtao Lu, Chunjing Xu
In British Machine Vision Conference, 2021
- Solipsistic Reinforcement Learning** ICLR 2021 Workshop
Mingtian Zhang*, Peter Noel Hayes*, Tim Z. Xiao, **Andi Zhang**, David Barber
In Self-Supervision for Reinforcement Learning Workshop - International Conference on Learning Representations, 2021

Honors & Awards

- Top Reviewer (10%) of AISTATS 2023** Feb. 2023
- The Williams/Kilburn Medal for outstanding final year student** Jul. 2017
School of Computer Science, University of Manchester
The medal is awarded on an occasional basis to students of exceptional distinction whose performance throughout their undergraduate course has been of outstanding merit.
- University awards for outstanding academic achievement** Jul. 2017
University of Manchester
The university will consider the top 0.5% of undergraduates (nominated by each school) for this award.

*Equal contribution.